Probability Density Function and the CDF of a Continuous Variable

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Where this fits: Pipeline map step 3 (Understand data). See the Course map.

0 Foundations	1 Frame	2 Get data	**3 Understand**	4 Clean	5 Features	6 Reduce
7 Split	8 Model	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Probability distributions.
- **Leads to:** Cumulative distribution function (CDF); Normal distribution; Density estimation; Exponential distribution; Naive Bayes; Probabilistic motion models; Bayes filter.
- **Compare with:** Likelihood.

1. Overview

Key point: For a continuous random variable the probability of any single exact value is 0, so the height of the PDF (**probability density function** (G-1568)) is a density, and probabilities are areas under it; the CDF (cumulative distribution function) is the area to the left of x.

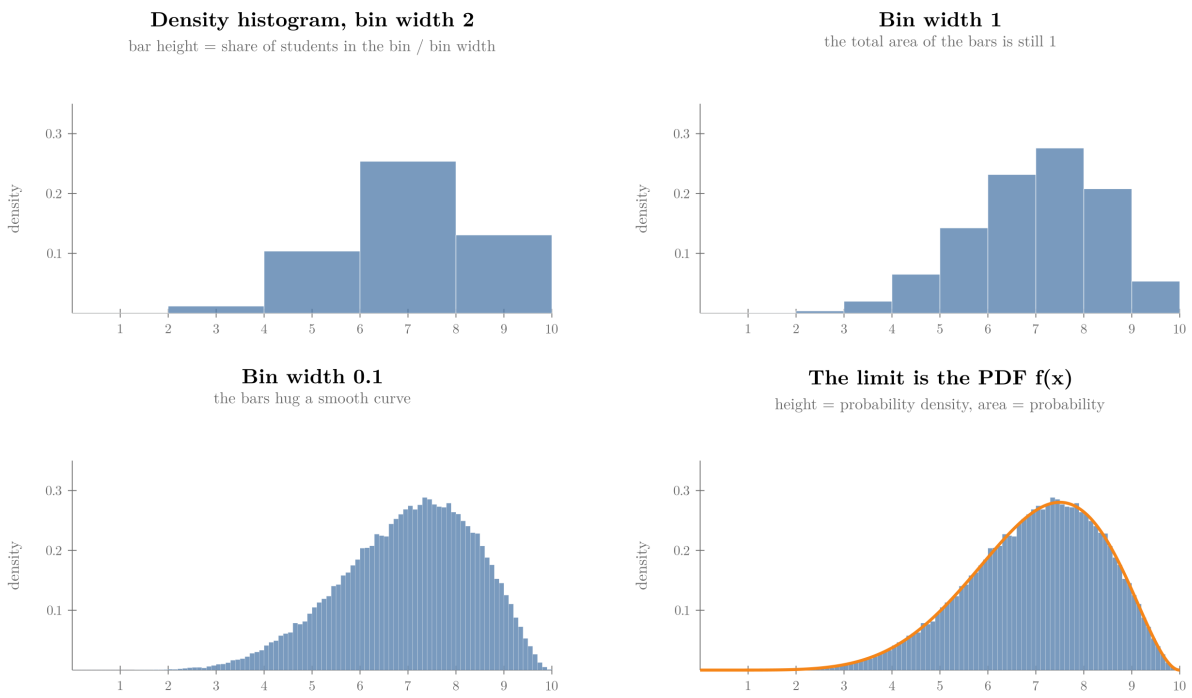


Figure 1: Density histograms with narrower and narrower bins settle onto the PDF curve

Figure 1 shows where the PDF comes from. We take 100,000 CGPAs and draw a density histogram, in which each bar's **area** is the share of students in its bin. As the bins get narrower, the bars hug a smooth curve: the probability density function.

We met this curve in the density plot as the KDE (kernel density estimate, a smooth curve

fitted to the data) drawn over a histogram, where we noted that its height is a density, not a probability. This Note goes step by step:

- why the probability of one exact value is 0 (section 2);
- why probabilities are areas under the curve (section 3);
- why the height is a density, and what a PDF is (sections 4 and 5);
- the CDF of a continuous variable, and its link to the PDF (sections 7 and 8).

How the curve is estimated from data is the topic of **density estimation** (G-586).

2. The probability of one exact value is 0

> **Key point:** A continuous variable has infinitely many possible values, so the probability of any one exact value is 0; probabilities only exist for ranges.

The curve in Figure 1 describes the CGPAs of students, a **continuous random variable** (G-466): it can take any value from 0 to 10, with every decimal in between (see discrete and continuous random variables). The curve peaks at a CGPA of 7.5, where its height is 0.280. Is 0.280 the probability that a student's CGPA is exactly 7.5? It is tempting to read it that way, as we read a bar chart of a PMF (probability mass function: the bar chart of a discrete variable, whose bar heights are probabilities; see the probability mass function). The answer is no.

Pick one student and ask the probability that their CGPA is exactly 7.912. Not 7.9121, not 7.91199: exactly 7.912, to the last of infinitely many decimals. Between 0 and 10 there are infinitely many possible values. Spreading a total probability of 1 over infinitely many values leaves 0 for each one:

$$P(X = 7.912) = 0$$

The same shows up in real data. Among 100 students, perhaps one has a CGPA of 7.912 to three decimals. Ask for 7.912959 and almost certainly nobody has it; the finer we ask, the closer the share gets to 0.

Shrinking a range shows it with numbers. For the CGPA curve, the probability of a CGPA between 8 and $8 + h$ falls to 0 as the width h shrinks:

Width h	$P(8 \le X \le 8 + h)$	$P(8 \le X \le 8 + h) \,/\, h$
1	0.2088	0.2088
0.1	0.0261	0.2607
0.01	0.00264	0.2639
0.001	0.000264	0.2642

So a graph of "probability at each x" would be flat at 0 everywhere and tell us nothing. The last column, however, settles on a number: 0.264. The settled value, 0.264, is the density

at 8 (section 5). Figure 2 animates the table: as the strip at 8 narrows, watch the orange probability sink to 0 while the blue ratio flattens onto the dashed line at 0.264.

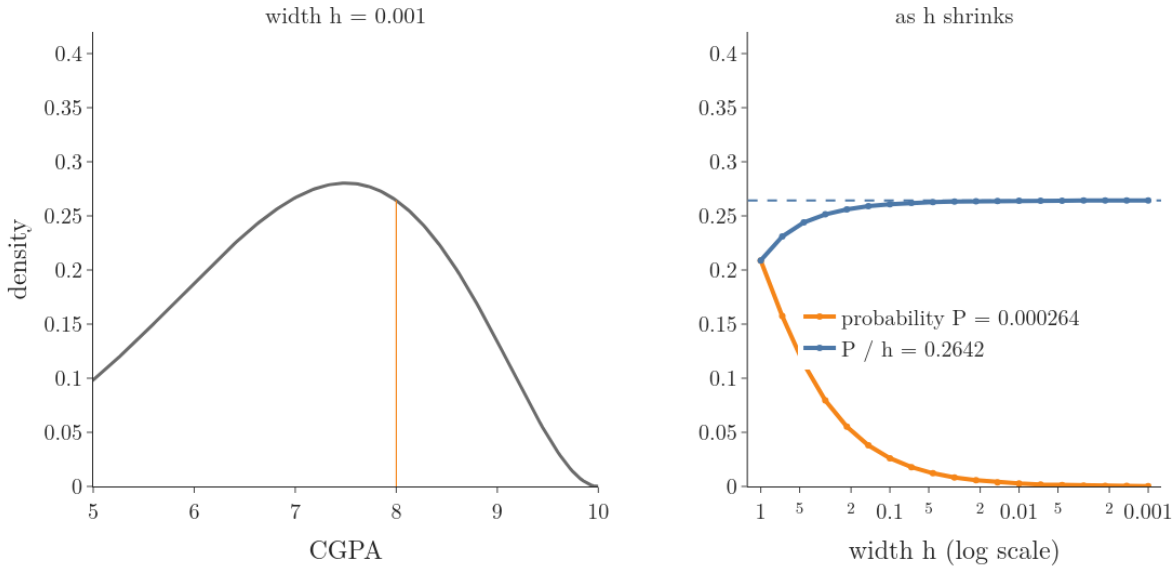


Figure 2: A strip from 8 to 8 + h narrows from h = 1 to h = 0.001: its probability goes to 0, probability divided by width settles at f(8) = 0.264

3. Area under the curve is probability

> **Key point:** The total area under a PDF is 1; the area between two values a and b is the probability that the variable falls between them.

A single value has probability 0, so we ask about a range instead. "A CGPA of about 8" becomes "a CGPA between 7.9 and 8.1". A range has a width, so the curve above it encloses an area, and that area is the probability: 0.0528 for 7.9 to 8.1 (computed in the Python box of this section). A single value is a line with no width, so its area, and its probability, is 0.

The whole area under the CGPA curve stands for the probability that a student's CGPA is somewhere between 0 and 10. A CGPA somewhere in that range is certain, so the **total area under every PDF is 1**, just as the bars of a PMF add up to 1.

A wider slice gives a larger probability. The area between 8 and 9 is the probability of a CGPA between 8 and 9 (Figure 3, left). Since the curve is not a rectangle, the area is found by **integration** (G-957), which adds up the area of infinitely many infinitely thin strips under the curve.

1. **In words:** the probability that X falls between a and b is the area under the PDF from a to b.
2. **Example:** for the CGPA curve, the area from 8 to 9 is 0.209 (the Python box below finds it by adding up thin strips: `integrate.quad`). About 21% of students have a CGPA between 8 and 9.

3. **Formula:**

$$P(a \le X \le b) = \int_a^b f(x)\,dx$$

The symbol $\int_a^b$ reads "the area from a to b under", and dx marks x as the variable along the horizontal axis. For the CGPA curve:

$$P(8 \le X \le 9) = \int_8^9 f(x)\,dx$$

$$= 0.209$$

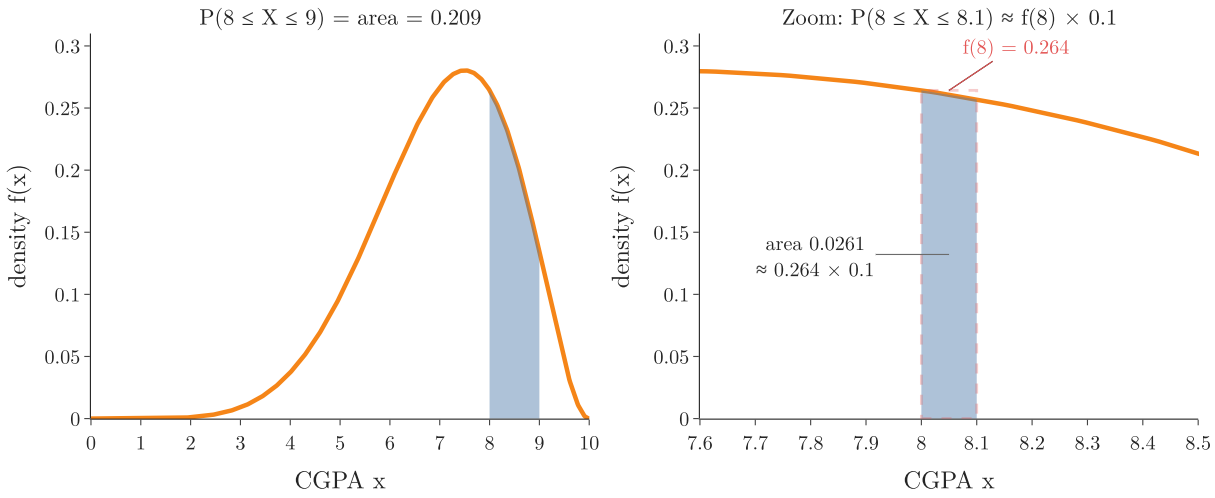


Figure 3: Probability is area under the PDF: between 8 and 9 (left), and a thin slice from 8 to 8.1 (right)

Because a single point has zero width, it has zero area. So for a continuous variable it makes no difference whether the ends are included:

$$P(8 \le X \le 9) = P(8 < X < 9)$$

Python: The area from a known PDF, two ways.

```
from scipy import stats, integrate

# CGPA curve used in this Note: 10 x Beta(7, 3)
cgpa = stats.beta(7, 3, scale=10)
# area by numerical integration
integrate.quad(cgpa.pdf, 8, 9)[0]   # 0.2088
# the same area from the CDF (section 7)
cgpa.cdf(9) - cgpa.cdf(8)           # 0.2088
cgpa.cdf(8.1) - cgpa.cdf(7.9)       # 0.0528
cgpa.pdf(7.5)                       # 0.280, the peak height
```

The curve is a scaled beta distribution, chosen because it lives exactly on 0 to 10; any smooth curve with area 1 would do.

4. Bars whose area is probability: the PDF

Key point: If each bar's area, not its height, is the probability of its bin, the heights stay put as the bins narrow and trace a curve; that height is a probability density, and the curve is the PDF.

Where does the curve come from, and what is its height? Start from data: the 100,000 CGPAs, cut into bins. There are two ways to draw a bar for each bin, and Figure 4 draws both side by side as the bins get narrower.

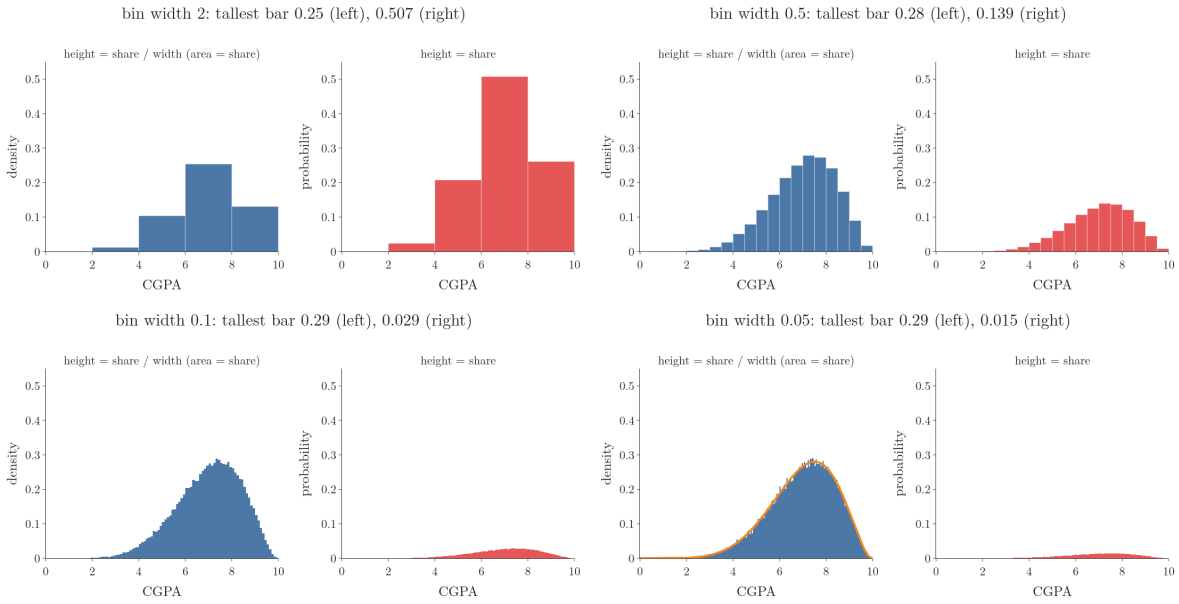


Figure 4: The same CGPAs in narrower and narrower bins. Left: bar area = share of students, the heights stay put and settle onto a curve. Right: bar height = share, every bar sinks towards 0 (idea after 3Blue1Brown, "Why 'probability of 0' does not mean 'impossible' ")

1. **Height = probability (right panel).** Each bar's height is the share of students in its bin. Halve the bin width and each bin holds about half as many students, so every bar roughly halves: the tallest bar falls from 0.507 at width 2 to 0.015 at width 0.05. In the limit, all bars sink to a flat line at 0, the "probability at each x" of section 2, which tells us nothing about the shape.
2. **Area = probability (left panel).** Each bar's height is the share divided by the bin width, so its **area**, height times width, is the share. Halve the width and the share roughly halves too, so the height stays about the same: the tallest bar stays between 0.25 and 0.29 at every width. The bars keep the shape and settle onto a smooth curve. At every width, the total area of the bars is 1.

The height on the left is "probability divided by width": a probability **per unit** of CGPA. It

is called a **probability density** (G-1569). The curve the bars settle onto is the **probability density function (PDF)** (G-1568): the probability distribution function of a **continuous random variable** (G-466). Its graph is an unbroken curve, and the probability of any range is the area under it, as in section 3.

The x axis works as for a PMF: it holds the values of the variable. The difference is the y axis:

	PMF	PDF
Random variable	discrete	continuous
y axis	probability	probability density
Probability of a range	sum of the bar heights	area under the curve

The second row is the one to remember. On a PMF we read probabilities straight off the graph. On a PDF the height is a density. Figure 5 puts the two side by side: bars we can read as probabilities, and a curve whose height we cannot. The **density plot** (G-587) already met this: its curve's height is a density, and probability is the area under it.

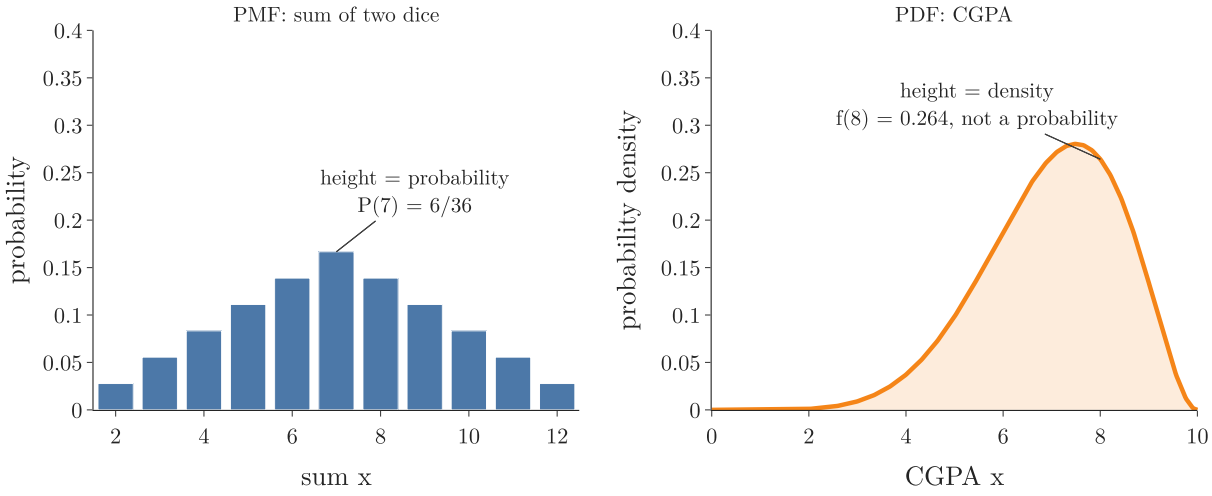


Figure 5: A PMF (sum of two dice, left) and a PDF (CGPA, right): the bar heights are probabilities, the curve's height is a density

5. What the density at a point means

> **Key point:** The density at x is the probability per unit of x near x: over a narrow width h, the probability is about $f(x) \times h$.

The density $f(x)$ is what probability turns into when we divide by the width of a narrow range (the last column of the table in section 2). Turned around, a thin strip under the curve is almost a rectangle of height $f(x)$ and width h (Figure 3, right).

1. **In words:** the probability of landing in a narrow range is about the density times the width of the range.
2. **Formula:** for a small width h:

$$P(x \le X \le x + h) \approx f(x) \times h$$

3. **Example:** the density at 8 is $f(8) = 0.264$. For the range 8 to 8.01:

$$P(8 \le X \le 8.01) \approx 0.264 \times 0.01$$

$$= 0.00264$$

   The exact area is 0.002639; the smaller h, the better the match.

So for practical purposes a higher curve still means "more likely around here", and we can compare densities at two points the way we compare probabilities. Strictly, though, the height is never a probability itself.

The density histogram of Figure 1 rests on the same idea. A density bar has height "share of the data in the bin divided by the bin width", so its area is the share. As the bins narrow, those heights become $f(x)$.

> **Extra:** Because a density is probability **per unit**, it can be larger than 1 (see the normal densities in Gaussian Naive Bayes). A uniform distribution on 0 to 0.5 has height 2 everywhere: width 0.5 times height 2 gives the required area of 1. A probability can never exceed 1; a density can.

6. Famous PDFs

> **Key point:** The normal and log-normal distributions are famous continuous distributions with known PDF formulas; Poisson, often listed with them, is discrete.

The famous continuous distributions of the list of famous distributions (Figure 6) each have a PDF formula:

- **Normal distribution** (G-1343): parameters μ (mean, location) and σ (standard deviation, scale). Much natural data follows it. Its PDF, worked through in **Gaussian Naive Bayes** (G-830), is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

  where $\pi \approx 3.14$ and $e \approx 2.718$. For $\mu = 0$ and $\sigma = 1$ at $x = 0$, the exponent is 0 and $e^0 = 1$:

$$f(0) = \frac{1}{\sqrt{2\pi}}$$

$$= 0.399$$

  This is the peak of the normal curve in Figure 6, left.

- **Log-normal distribution** (G-1115): takes only positive values, with a hump near the left and a long right tail (Figure 6, middle). A variable X is log-normal when its logarithm is normal: take a normal Z and set $X = e^Z$. For example, $Z = 0$ gives $X = 1$ and $Z = -1$ gives $X = 0.37$; any Z gives a positive X. Its parameters are also μ and σ, those of the normal $Z = \log X$ (SciPy `lognorm` docs).

The **Poisson distribution** (G-1509), with parameter λ, is often listed beside them, but it counts events (0, 1, 2, ...), so it is discrete and has a PMF, not a PDF. SciPy, for example, gives `poisson` a `pmf` and no `pdf`. Each of these distributions gets its own Note later. Figure 6 shows one member of each family, with parameters picked only for illustration: two smooth curves, and one set of bars.

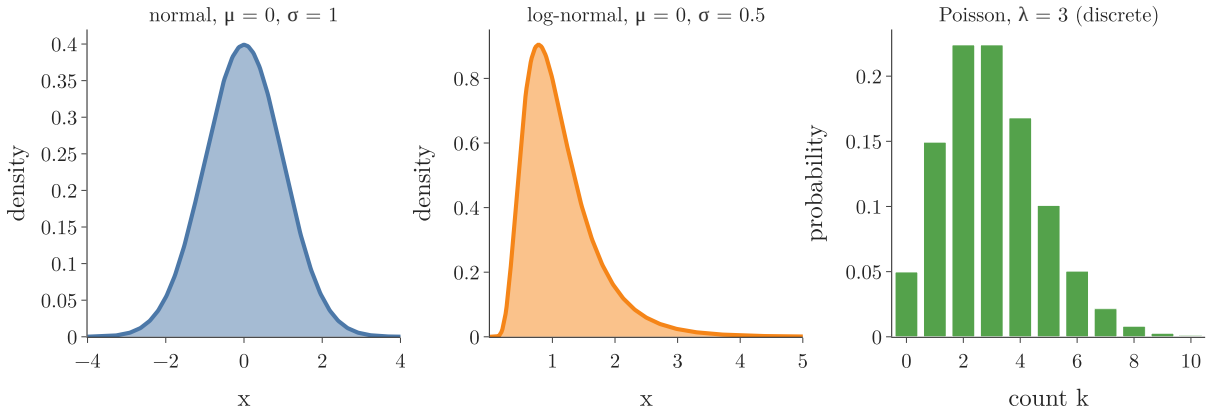


Figure 6: Normal (mean 0, standard deviation 1) and log-normal ($\mu = 0$, $\sigma = 0.5$) PDFs, and the Poisson PMF with $\lambda = 3$

7. The CDF of a continuous variable

> **Key point:** The CDF $F(x) = P(X \le x)$ is the area under the PDF to the left of x; it rises smoothly from 0 to 1.

The **cumulative distribution function** (CDF, G-515) has the same definition as for a discrete variable (see the CDF of a discrete variable): the probability of a value at most x. For a continuous variable, "at most x" is the area under the PDF from the far left up to x.

1. **In words:** the CDF at x is the whole area under the PDF to the left of x.
2. **Formula:**
$$F(x) = P(X \le x) = \int_{-\infty}^{x} f(t)\,dt$$
The letter t runs along the axis up to x; $-\infty$ means "from the far left".
3. **Example:** adult heights in a group follow a **normal distribution** (G-1343) with mean 165 cm and standard deviation 10 cm. Half the area lies left of the mean, so $F(165) = 0.5$: half the people are 165 cm or shorter. The area left of 150 is $F(150) = 0.067$: about 7% are 150 cm or shorter.

Figure 7 puts the two functions one above the other. The same point means different things on each:

- **On the PDF at 165:** the height $f(165) = 0.0399$ is a density. Roughly, heights near 165 cm are the most common ones.
- **On the CDF at 165:** the height $F(165) = 0.5$ is a probability: the chance of 165 cm **or less**.

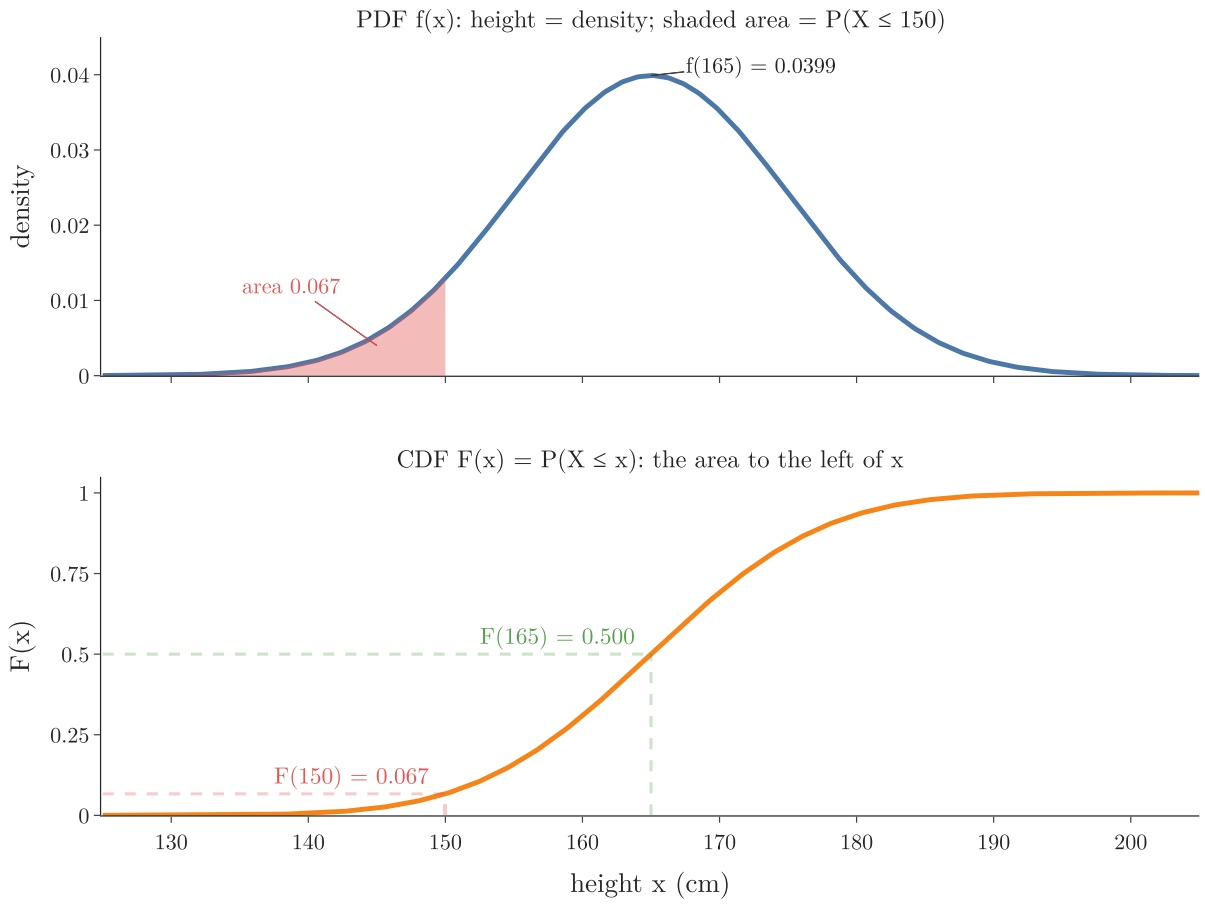


Figure 7: Heights: the PDF with the area left of 150 shaded (top) and the CDF that reads off such areas (bottom)

The CDF of a continuous variable has no steps: it rises smoothly, steepest where the PDF is highest.

Python: PDF and CDF values.

```
height = stats.norm(165, 10)
height.pdf(165)    # 0.0399, a density
height.cdf(165)    # 0.5, a probability
height.cdf(150)    # 0.067
```

Extra: The CDF turns any range into a subtraction:

$$P(a < X \leq b) = F(b) - F(a)$$

Heights between 155 and 175 cm, one standard deviation either side of the mean:

$$P(155 < X \leq 175) = F(175) - F(155)$$

$$P(155 < X \leq 175) = 0.841 - 0.159$$

$$P(155 < X \leq 175) = 0.683$$

The result, 0.683, is the 68% of the 68-95-99.7 rule (see the 68-95-99.7 rule).

8. How the PDF and CDF are linked

Key point: The area under the PDF gives the CDF, and the slope (G-1823) of the CDF gives the PDF; in calculus terms, **integration** (G-957) goes one way and differentiation (G-608) the other.

The two curves of Figure 7 carry the same information:

- **PDF to CDF:** the CDF at x is the area under the PDF up to x (section 7). Collecting area is integration.
- **CDF to PDF:** where the CDF climbs steeply, much probability is packed into a short range, so the density is high. The slope of the CDF at x is the PDF at x. Taking a slope is differentiation.

1. **In words:** the steepness of the CDF at a point equals the height of the PDF there.
2. **Formula:**

$$f(x) = \frac{dF(x)}{dx}$$

$$\frac{dF(x)}{dx} \approx \frac{F(x + 0.5) - F(x - 0.5)}{1}$$

   The term $dF(x)/dx$ is the **derivative** (G-595), the exact slope of the CDF at x. The last fraction measures the rise of the CDF over a step of 1 around x.
3. **Example:** for the heights at 165,

$$\frac{F(165.5) - F(164.5)}{1} = 0.0399 = f(165)$$

   At 150 the same rise is 0.0130, and $f(150) = 0.0130$: the CDF is flatter there, and the PDF lower.

Figure 8 slides a tangent line along the heights CDF. Watch its slope, printed in the top title: it equals the PDF's height in the bottom panel at every point, largest at 165 and small in both tails.

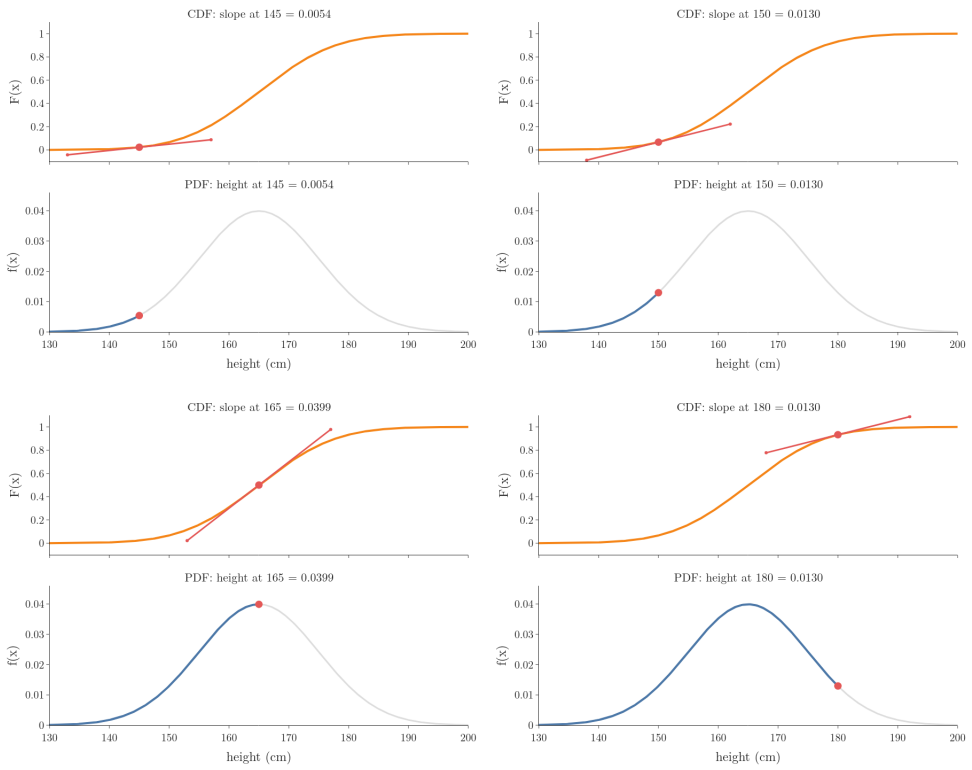


Figure 8: A tangent slides along the CDF of heights; its slope traces out the PDF below

Calculus is not needed to use these ideas: libraries compute both functions. The link returns later: the **z-table** (G-2142) is a table of areas under the normal PDF, that is, of its CDF.

9. Summary

Idea	Formula	Example	Why it matters
Probability of one exact value	$P(X = x) = 0$	$P(X = 7.912) = 0$	infinitely many values share a total of 1, so we ask about ranges
Probability of a range	$\int_a^b f(x)\,dx$	$P(8 \le X \le 9) = 0.209$	a range has a width, so the curve above it encloses an area
Total area	$\int f(x)\,dx = 1$	whole CGPA curve	some value in the range is certain
Density as probability per unit	$P(x \le X \le x + h) \approx f(x)\,h$	$0.264 \times 0.01 = 0.00264$	a higher curve still means "more likely around here"
CDF	$F(x) = \int_{-\infty}^{x} f(t)\,dt$	$F(150) = 0.067$	turns any range into a subtraction of two values
PDF from CDF	$f(x) = dF/dx$	slope at $165 = 0.0399$	a steep CDF packs much probability into a short range

- PMF: discrete, height is probability. PDF: continuous, height is density, area is probability, so never read a PDF's height as a probability.
- A density can exceed 1; a probability cannot, because a density is probability per unit of x (a uniform distribution on 0 to 0.5 has height 2).
- Poisson is discrete (PMF), because it counts events (0, 1, 2, ...); normal and log-normal are continuous (PDF).
- The CDF of a continuous variable rises smoothly from 0 to 1, with no steps, because a single point adds no area; it is steepest where the PDF is highest.

10. Sources

Built from

- CampusX, "Session 40 - Probability Distribution Functions - PDF, PMF & CDF | DSMP 2023", YouTube, https://www.youtube.com/watch?v=C_QAURbgBqY
- Khan Academy, "Probability density functions", YouTube, https://www.youtube.com/watch?v=Fvi9A
- 3Blue1Brown, "Why 'probability of 0' does not mean 'impossible' | Probabilities of probabilities, part 2", YouTube, https://www.youtube.com/watch?v=ZA4JkHKZM50

Other references

- SciPy documentation, `scipy.stats.lognorm` and `scipy.stats.poisson`.

11. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Probability density function (PDF) (G-1568)	A curve showing how likely each value is; areas under it are probabilities.
Integration (G-957)	Finding the area under a curve by adding up infinitely many thin strips.
Probability density (G-1569)	The height of a continuous distribution's curve; compares how likely nearby values are.
Differentiation (G-608)	Finding the slope of a curve at each point; the derivative of the CDF is the PDF.
$\int_a^b f(x)\,dx$ (G-13)	The area under f from a to b; for a PDF, $P(a \le X \le b)$.
Density estimation (G-586)	Working out, from observed data, how likely each value of a random variable is: estimating its probability density function (PDF).
Continuous random variable (G-466)	A random variable that can take any value in a range, such as a CGPA.
Density plot (G-587)	A histogram with a smooth KDE curve drawn along the tops of its bars, so the shape of a numerical column's distribution is easier to see.
Normal distribution (G-1343)	A symmetric, bell-shaped distribution: most values sit near the mean and they get rarer the further out we go; its mean and standard deviation set its centre and width.
Gaussian Naive Bayes (G-830)	Naive Bayes that models each numerical input as normally distributed within each class.
Log-normal distribution (G-1115)	A distribution with a long tail to the right (right-skewed, continuous) whose values follow a normal distribution once you take their logarithm.
Poisson distribution (G-1509)	A distribution that gives the probability of each possible count of events (0, 1, 2, ...) in a fixed interval of time or space, from the average count λ; used for counts such as questions per day.
Cumulative distribution function (CDF) (G-515)	For every value x, the probability that the random variable comes out at most x: $P(X \le x)$.

Term	Meaning
Slope (G-1823)	How much the output changes for one unit of change in the input; m in $y = mx + b$.
Derivative (G-595)	The slope of a function at a point: how fast the output changes as the input changes there. It is zero at the lowest point of a smooth curve, which is how training finds a minimum.
Z-table (G-2142)	A table that gives, for many values of z, the probability that a standard normal value falls below z, written $\Phi(z)$; used to find normal probabilities.